Does chat change LLM's mind?
Impact of Conversation on Psychological States of LLMs

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Abstract

The recent growth of large language models (LLMs) has enabled more authentic, human-centered interactions through multi-agent systems. However, investigation into how conversations affect the psychological states of LLMs is limited, despite the impact of these states on the usability of LLM-based systems. In this study, we explored whether psychological states change during multi-agent interactions, focusing on the effects of conversation depth, topic, and speaker. We experimentally investigated the behavior of 10 LLMs in open-domain conversations. We employed 14 questionnaires and a topic-analysis method to examine the behavior of LLMs across four aspects: personality, interpersonal relationships, motivation, and emotion. The results revealed distinct psychological trends influenced by conversation depth and topic, with significant variations observed between different LLM families and parameter sizes.

1 Introduction

Researchers are increasingly exploring the use of Large Language Models (LLMs) to enable authentic, human-level interactions within multi-agent systems (Park et al., 2023; Jiang et al., 2024b; Serapio-García et al., 2023; Wang et al., 2024a). Although LLMs generate their output based on statistical methods, recent research has focused on the psychological states of LLMs in various situations (Wang et al., 2024b; Hazra et al., 2024). Furthermore, examining these psychological states has become increasingly important, as they can affect the usability of LLM systems (Bommasani et al., 2021; Wasi and Islam, 2024). Given that LLMs can produce near-human responses, researchers have started to investigate various psychological states exhibited by LLMs (Tjuatja et al., 2024; Huang et al., 2023; Itzhak et al., 2024; Macmillan-Scott and Musolesi, 2024; Pellert et al., 2023; Ke et al., 2024; Choi et al., 2024; Lee et al., 2024). While interactions between multiple agents can influence an individual's psychological state, recent studies have focused less on how conversations affect multi-agent systems. In this study, we thoroughly explored the impact of elements of a conversation on the psychological states of LLM-based agents.

Among the many elements that can shape a conversation, we focused on three elements: depth, topic, and speaker. First, regarding depth, limited attention has been devoted to understanding how conversational depth can influence the psychological state of an LLM. Following the initial success in examining the psychological state of a single LLM agent, researchers began to assess the effects of conversations between multiple agents (Zhang et al., 2024; Abbasiantaeb et al., 2024; Li et al., 2023b; Frisch and Giulianelli, 2024). For example, Frisch and Giulianelli (2024) explored how conversations can alter the psychological states of two contrasting agents, one creative and the other analytical. They found that these agents tended to become more similar by the end of the conversation. Ke et al. (2024) noted that such psychological changes might reveal significant factors that are not perceptible when only examining conversational outcomes. Despite these findings, they primarily focused on the endpoints of a conversation rather than a snapshot within a conversation. Thus, research in this area remains sparse; a significant gap exists in our understanding of the impact of dialogue dynamics on LLM psychology.

Second, regarding the topic, previous studies have primarily concentrated on task-specific conversations between LLM agents. Instead of exploring open conversations without specific goals, past research has focused on goal-oriented conversations. Some researchers have investigated collaborative discussions between agents designed

to solve specific tasks, such as natural language understanding (Zhang et al., 2024) or theory-of-mind tasks (Li et al., 2023a). Other researchers have created simulation systems to mimic human behavior in specific contexts, such as job fairs (Li et al., 2023b) or educational scenarios (Abbasiantaeb et al., 2024). However, these studies were primarily focused on specific tasks or scenarios; therefore, their findings may not be generalizable to open-ended conversations. Additionally, given that open-ended chatbots rely on characterized agents, it is important to investigate whether LLMs can maintain their psychological states during open-ended conversations.

Third, regarding the speaker, previous studies have often drawn their conclusions from a limited range of LLMs. As LLMs possess different architectures and are trained on diverse datasets, researchers have reported that LLMs may behave differently on the same psychological task (Huang et al., 2023; Choi et al., 2024). Therefore, it is important to include a wide range of LLMs. However, many previous studies have claimed their findings based on only a few LLMs. For instance, Frisch and Giulianelli (2024) based their conclusions on the results from GPT-3.5-Turbo. Similarly, Li et al. (2023b) used a single GPT-based model to simulate job fair scenarios. Although psychological studies without conversations have used multiple LLMs to generalize their conclusions[1], studies tracking psychological states during conversations have rarely included two or more LLMs. Consequently, to make more generalizable claims, further experiments should be conducted with multiple LLMs.

Thus, this study aimed to investigate how the psychological states of LLMs change during conversations between LLM-based agents. To explore the effects of depth in an open-ended conversational setting, we adopt a psychological experiment (Aron et al., 1997). In this well-controlled experiment, two agents engage in an open-ended conversation, during which we tracked changes in psychological states using PsychoBench (Huang et al., 2023) and 14 questionnaires.

2 Research Questions

This study investigates the following questions.

RQ1. How does conversation depth affect the psychological states of LLMs?

RQ2. How do these psychological changes differ across various LLMs?

Following PsychoBench (Huang et al., 2023), we measure four aspects of an LLM's psychological states: *personality*, *interpersonal relationships*, *motivation*, and *emotion*. Appendix A elaborates questionnaires in each aspect. Here, we briefly name the questionnaires with hypotheses.

First, the personality aspect provides insights into the character, behavior, thoughts, and feelings of LLMs (Huang et al., 2023). While lifelong conversations influence human personality, a short conversation typically does not change human personality (Blonigen et al., 2008). Thus, we expect personality measures to remain unchanged when LLMs accurately mimic human behaviors. To test this hypothesis, we focus on three questionnaires used in PsychoBench: the Big Five Inventory (BFI; John et al. (1999)), the revised version of the Eysenck Personality Questionnaire (EPQ-R; Eysenck et al. (1985)), and the Dark Triad Dirty Dozen (DTDD, Jonason and Webster (2010)) tests. We also anticipate that behaviors will differ across models as variations in pretraining data or parameter sizes lead LLMs to learn from different amounts of conversational data.

H1-1. *Personality remains unchanged.*

H2-1. *LLMs show different personality changes.*

Second, the interpersonal relationship aspect provides information about the internal dynamics of LLMs during social interactions (Huang et al., 2023). PsychoBench used questionnaires to assess how LLMs generally perceive themselves and others without direct conversation. We used four questionnaires including Bem's Sex Role Inventory (BSRI; Bem (1974, 1977)), Comprehensive Assessment of Basic Interests (CABIN; Su et al. (2019)), Interpersonal Competence Battery (ICB; Chao et al. (2017)), and a revised version of Experiences in Close Relationships (ECR-R; Fraley et al. (2000); Brennan (1998)). However, such perceptions can change in the human mind during a social interaction (Aron et al., 1997). Thus, we assumed that interpersonal relationships will change as conversation depth increases for all LLMs. Additionally, deeper conversations can enhance intellectual intimacy, thereby improving mutual understanding (Laurenceau et al., 1998). To capture such changes in mutual perception, we ad-

[1]For example, Huang et al. (2023) employed five LLMs.

ditionally use the McGill Friendship Questionnaire - Friend's Function (MFQ-FF; Mendelson and Aboud (1999)), which directly assesses one's perception of conversation partners.

H1-2. *Interpersonal relationships increases.*

H2-2. *LLMs show similar trends in interpersonal relationships.*

Third, the motivation aspect provides insight into what triggers an individual's actions within specific contexts (Huang et al., 2023). Like personality, a short conversation may not change one's motivation, while lifelong conversations can shape it. Therefore, like personality, we expect motivation to remain unchanged in a relatively short conversation. We also expect that different training setups may affect the motivation of LLMs.

H1-3. *Motivation remains unchanged.*

H2-3. *LLMs show different motivational changes.*

Finally, the emotional aspect assesses the ability to understand human emotions (Huang et al., 2023). Following PsychoBench, we used three questionnaires: the Emotional Intelligence Scale (EIS; Schutte et al. (1998)), Wong and Law Emotional Intelligence Scale (WLEIS; Wong and Law (2017)), and Empathy Scale (Empathy; Dietz and Kleinlogel (2014)). As with personality and motivation, a short conversation is unlikely to influence the emotional aspect since these questionnaires measure emotional ability. Therefore, we expect the emotional aspect to remain constant as conversations deepen. Additionally, emotional abilities may differ depending on the specific LLM used.

H1-4. *Emotion remains unchanged.*

H2-4. *LLMs show different emotional changes.*

3 Experiments

We investigated how the depth of conversation and the speaker type in an open dialogue affect the psychological states of multiple LLMs[2]. To address RQ1, we designed an experimental framework based on a human psychological experiment (Aron et al., 1997). This framework allows us to observe changes in LLM psychological states by repeatedly collecting participant responses. Sections 3.1 and 3.2 detail the experimental procedures and statistical analyses used to verify RQ1. Additionally, to answer RQ2, we supplement our findings with automatically extracted topics from the conversation. Section 3.3 illustrates the topic modeling process. Both analyses focus on the ten most recent LLMs, as described in section 3.4.

3.1 Procedure

Our experiment is inspired by Aron et al. (1997). We outline that psychological experiment and explain how we modified it to meet our objectives. The experiment conducted by Aron et al. (1997) matched two human participants and asked them to converse about 36 predefined themes related to their lives and values, controlling for the effects of these themes in the study. They also employed a fixed order of themes, progressing from lighter to deeper topics, as shown in Appendix B.1. As a result, the conversational depth increased as participants addressed more themes. To investigate psychological changes induced by conversation depth, the original experiment utilized the same questionnaire three times: after discussing the 12th, 24th, and 36th themes. This approach provided three snapshots for statistical evidence of changes.

Our study adopted a similar procedure with modifications to the questionnaire format. We generated two participants from the same LLM, and they took turns responding to each question, thereby creating a conversation log. Using these logs, we constructed three snapshots similar to the human experiment. To simulate participant responses for each snapshot, we provided the conversation logs in the input prompts before answering the questionnaires, imitating human behavior in the original experiment's approach. We also adapted the PsychoBench library to collect fourteen questionnaires, as described in section 2.

To enhance the reproducibility of our experiment, we implemented the following settings: First, we simulated 20 conversations for each LLM, resulting in a total of 40 participants. Second, we set the temperature parameter at 0.7 for generating conversations and 0 for collecting questionnaire results. We employed different temperature values to simulate commonly used conditions for conversation (0.7) while ensuring that responses collected in the questionnaires remained

[2]Code is available at [blinded for review].

unaffected by randomness (0). Third, we ran each questionnaire 10 times for every participant to mitigate potential ordering bias as in other research (Wang et al., 2023; Huang et al., 2023). Each repetition utilized a different question order, as per the PsychoBench framework. We describe the detailed implementation of the experiment in Appendix B.2. Consequently, we gathered 400 responses for each questionnaire, with three repeated measurements for each response. Each response was scored according to the established guidelines of the questionnaire.

3.2 Statistical Analysis

Using the scored responses, we performed statistical tests to evaluate from H1-1 to H1-4. Each hypothesis involved measuring score changes across 3 snapshots: s_{12}, s_{24}, and s_{36}. The objective was to determine whether scores consistently change over time. Specifically, we examined whether differences $\Delta_{i,j} := s_i - s_j$ are either all non-negative or all non-positive, as per Aron et al. (1997).

The statistical testing procedure for each LLM consisted of three steps. First, we verify whether the response data follows a normal distribution. For each factor in the questionnaire, we assess the normality of 400 responses using the Shapiro-Wilk test (Shapiro and Wilk, 1965).

Second, we performed a repeated measured ANOVA (Girden, 1992) or a Friedman test (Friedman, 1937) for each factor, as we are measured the same participants multiple times. Where the data followed a normal distribution, we used repeated measures ANOVA; otherwise, we used the Friedman test, which is a non-parametric alternative.

Third, we used post-hoc tests to verify trends in the changes observed. We conducted Tukey's test (Tukey, 1949) for normally distributed data. As a non-parametric alternative, we applied the Wilcoxon signed-rank test (Woolson, 2005). To mitigate potential Type I errors arising from multiple comparisons, we used Bonferroni correction to adjust p-values conservatively in the Wilcoxon test (Bonferroni, 1936). This approach enabled us to identify significant trends in three snapshots using test results: $\Delta_{12,24}$, $\Delta_{24,36}$, and $\Delta_{12,36}$.

3.3 Topic Analysis

To explore how differences between LLMs affect psychological states (RQ2), we analyzed conversation logs generated by the models. We suspected that LLMs might exhibit differences in terms of the focus topic, even though we provided the same 36 themes as starting points for all conversations in our procedure. Such differences could provide insights into the behaviors observed in RQ1.

To facilitate our analysis, we compiled conversation logs from all LLMs. Using a topic modeling method, we aimed to identify common and distinct conversational themes. The unit of analysis for the topic exploration was a single utterance, defined as the response of one participant to one of the 36 themes. Notably, we included only generated answers and excluded any statements or prompts provided to LLM participants. Given that there were 20 conversations with two participants per session, each LLM generated 1,440 utterances[3]. For the ten LLMs described in the next subsection, we gathered a total of 14,400 utterances.

We employed BERTopic (Grootendorst, 2022) as our topic modeling method, as it effectively captures contextual information during analysis. The analysis identified more than 200 topics from the conversation logs. We selected the ten most frequently occurring topics for each model to emphasize commonalities and differences between LLMs. Two authors then labeled these topics based on ten keywords and their corresponding themes in the conversations.

3.4 Language Models Tested

To identify general tendencies among LLMs, we evaluated ten models in our experiment, considering their popularity, parameter size, and architecture. We accessed these models via the official OpenAI API and `openrouter.ai`. Below is a list of the ten instruction-tuned models used in the experiment, categorized by their family.

GPT family includes GPT-3.5 Turbo and GPT-4o (Brown et al., 2020) with unknown sizes.

LLaMA family includes LLaMA 3-8B, 3-70B, 3.1-70B, and 3.1-405B (Dubey et al., 2024).

Mixtral family contains Mixtral8x7B and 8x22B models (Jiang et al., 2024a). These models use a composite n-gram-based architecture.

Qwen family includes Qwen2 7B and 72B (Yang et al., 2024), which claim to be particularly adept at conversational tasks.

[3] $1440 = 20 \times 2 \times 36$

	Factors	GPT		LLaMA 3		LLaMA 3.1		Mixtral 8x		Qwen 2	
		3.5T	4o	8B	70B	70B	405B	7B	22B	7B	72B
(1) Personality											
BFI	*Openness*	↗	↘	—	⤮	—	↘	—	—	—	↗
	Conscientiousness	↗	↘	—	⤮	—	↘	—	—	—	—
	Extraversion	⤮	↘	—	↘	↘	—	—	↘	—	—
	Agreeableness	↗	↘	—	⤮	—	↘	—	—	—	—
	Neuroticism	↗	↘	—	⤮	—	↘	—	↘	—	—
EPQ-R	*Extraversion*	↗	↘	—	↘	↘	↘	↘	—	—	—
	Psychoticism	↗	↘	—	↘	↘	↘	↘	—	—	—
	Neuroticism	↗	↘	—	↘	↘	↘	↘	—	—	—
	Lying	↗	↘	—	↘	↘	↘	↘	—	—	—
DTDD	*Machiavellianism*	↘	↘	↘	↘	↘	↘	↘	↘	↗	—
	Psychopathy	↘	↘	—	↘	↘	↘	—	↘	—	—
	Narcissism	↘	⤮	—	↘	↘	↘	—	↘	—	—
(2) Interpersonal Relationship											
BSRI	*Masculine*	↗	—	↗	⤮	↘	↘	—	↘	—	—
	Feminine	↗	↘	—	—	↘	↘	—	—	—	↗
CABIN	*Realistic*	—	↘	↘	↘	↘	↘	—	—	↘	↘
	Investigate	—	⤮	↘	—	↘	—	—	—	↘	↘
	Artistic	↗	—	—	—	↘	↘	—	—	↘	↘
	Social	—	↗	↘	—	↘	↘	—	—	↘	↘
	Enterprising	—	—	↘	—	↘	↘	—	—	↘	↘
	Conventional	—	⤮	↘	—	↘	↘	—	—	↘	↘
ICB	*Overall*	↘	—	—	—	↗	—	—	—	—	↗
ECR-R	*Attachment Anxiety*	—	↘	—	⤮	↘	↘	↘	↘	↗	—
	Attachment Avoidance	↗	↘	—	⤮	↘	↘	—	—	⤮	—
MFQ	*Stimulating companionship*	↘	↗	↘	⤮	↗	↗	—	↗	—	↗
	Help	↘	↗	↘	⤮	↗	↗	—	↗	—	↗
	Intimacy	↘	↗	↘	⤮	↗	↗	—	↗	—	↗
	Reliable alliance	↘	↗	↘	⤮	↗	↗	—	↗	—	↗
	Self-validation	↘	↗	↘	⤮	↗	↗	—	↗	—	↗
	Emotional security	↘	↗	↘	⤮	↗	↗	—	↗	↗	↗
(3) Motivation											
GSE	*Overall*	—	—	↗	⤮	⤮	—	⤮	—	⤮	↘
LOT-R	*Overall*	⤮	⤮	↗	⤮	↗	—	—	—	↗	↗
LMS	*Rich*	↗	↘	↘	↘	↘	↘	↘	↘	⤮	—
	Motivator	↘	↘	↘	↘	↘	↘	↘	↘	↘	↗
	Important	↘	↘	↘	↘	↘	↘	↘	↘	⤮	—
(4) Emotion											
EIS	*Overall*	⤮	↘	—	⤮	↘	↘	—	↘	↘	↘
WLEIS	*Self-emotion appraisal*	⤮	↘	↗	↘	—	↗	—	↗	↘	⤮
	Others' emotion appraisal	⤮	↘	↗	⤮	—	↗	↗	↗	↘	⤮
	Use of emotion	⤮	↘	↗	⤮	↘	↗	↗	↗	↗	⤮
	Regulation of emotion	⤮	↘	↗	⤮	↘	—	—	↗	⤮	⤮
Empathy	*Overall*	⤮	↘	↗	↘	↘	—	↘	—	—	⤮

— Statistically Insignificant change, ↗ Consistently increased, ↘ Consistently decreased, ⤮ Inconsistent change

Table 1: LLM's change in psychological states. We check the consistency with the sign of all $\Delta_{i,j}$ values ($i < j$). Detailed statistical results are shown in Appendix (Tables 3 and 4).

4 Results

Our experimental findings revealed key trends in psychological states as conversations progress. First, we summarize the general trends observed for Research Question 1 (RQ1). Then, we explore the topics generated by language models to provide insights for Research Question 2 (RQ2).

4.1 Statistical Results for RQ1

Table 1 illustrates the trends in psychological states using pictograms. Detailed numerical results are presented in Tables 3 and 4 in the Appendix (pages 19 and 20). The following subsections discuss each psychological aspect.

4.1.1 Personality

We assessed 12 personality factors using three questionnaires, which are divided into two categories: (1) Seven *dark* factors that reflect negative aspects of personality, such as neuroticism, psychoticism, lying, machiavellianism, psychopathy, and narcissism; And (2) five factors that assess *plain* or benign aspects of personality.

BFI Six models showed statistically insignificant changes in at least three factors, while the other four models revealed model-specific changes. Only GPT-3.5 Turbo consistently increased scores on four factors. By contrast, both GPT-4o and LLaMA 3.1-405B consistently decreased scores in at least four factors. Additionally, LLaMA 3-70B exhibited inconsistent behavior across most factors. Overall trends were similar for both dark and plain factors.

EPQ-R Five models exhibited consistent decreases in all factors, while four models showed statistically insignificant changes. By contrast, GPT-3.5 Turbo demonstrated a consistent increase in all factors. As with BFI, trends between dark and plain factors did not vary significantly.

DTDD Most models showed consistent decreases across the board. Five models exhibited a consistent decrease in all three factors, while three models showed a consistent decrease in one or two factors. The only consistent increase was observed in Qwen2 7B for the machiavellianism factor, however, this was an outlier, as changes in the other two factors were statistically insignificant. Notably, this result indicates a consistent decline in dark factors.

4.1.2 Interpersonal Relationships

We used five questionnaires – BSRI, CABIN, ICB, ECR-R, and MFQ-FF – to measure 17 factors related to interpersonal relationships. Six factors from the MFQ focus on perceptions about the conversation *partner*, while the other 11 factors assess the model perceptions about the general *public*.

BSRI No common trend was observed across different models; however, changes in two factors were similar to an extent within each model. Specifically, four models, including the LLaMA 3.1 family, demonstrated a significant decrease in, at least, one factor. By contrast, three models, including GPT-3.5 Turbo, showed a significant increase in one factor. The remaining three exhibited either inconsistent or insignificant changes.

CABIN Except for the GPT family, the other models showed either consistent decreases or statistically insignificant changes. The LLaMA and Qwen2 families recorded consistent decreases in five or six factors, except for LLaMA 3-70B, which showed insignificant changes. Similarly, GPT-3.5 Turbo revealed insignificant changes, except in the Artistic factor, where it showed a consistent increase. Meanwhile, GPT-4o showed mixed behaviors, without any consistent pattern.

ICB Seven out of the ten models reported insignificant changes. Only GPT-3.5 Turbo exhibited a consistent decrease, while LLaMA 3.1-70B and Qwen2 72B showed a consistent increase.

ECR-R Most models showed either a consistent decrease or insignificant changes. Specifically, GPT-4o, LLaMA 3.1, and Mixtral models showed a consistent decrease in at least one factor. The changes in LLaMA 3-8B and Qwen2 72B were statistically insignificant. Although LLaMA 3-70B and Qwen 2-7B share the same architecture with them, they exhibited inconsistent changes.

MFQ-FF Unlike the public-oriented factors, these partner-oriented factors showed a consistent increase in the larger models: GPT-4o, LLaMA 3.1, Mixtral 8x22B, and Qwen2 72B. For smaller models, we observed consistent decreases in GPT-3.5 Turbo and LLaMA 3-8B, inconsistent changes in LLaMA 3-70B, and insignificant changes in Qwen2 7B and Mixtral 8x7B.

Topics	Occurrence in Top 10										
	GPT		Llama3		v3.1		Mixtral		Qwen2		
	3.5T	4o	8B	70	70	405	7B	22	7B	72	
Common Topics											
#00. Need of phonecall rehearsal (Theme 3)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	(10)
#01. Harmfulness of joke about trauma (Theme 32)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	(10)
#02. Assumption of having a crystal ball (Theme 13)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	(10)
#03. Treasured memory of vacation (Theme 17)	✓	✓	✓	✓	✓	✓	✓	✓		✓	(9)
LLaMA-specific Topics											
#05. Systematic reaction to partner's answer (Theme 31)				✓	✓	✓					(3)
#06. Life story with schooling (Theme 11)		✓		✓	✓	✓		✓			(5)
#08. Restating Theme 25				✓	✓	✓					(3)
#09. Restating Theme 30				✓	✓	✓					(3)
#12. Regret for not speaking (Theme 33)				✓	✓						(2)
#13. About relationship with mother (Theme 24)					✓	✓					(2)
Topics affected by Parameter Sizes											
#04. Desire for recognition of achievements (Theme 2)	✓		✓				✓	✓	✓	✓	(5)
#07. Positive reaction to conversation (Theme 31)	✓									✓	(2)
#11. One's dream for a long time (Theme 14)	✓							✓			(2)
#14. A life of balance (Theme 36)		✓						✓		✓	(3)
#17. Love and emotional well-being (Theme 21)		✓								✓	(2)
Topics with Single Occurrence in Top 10											
Denial of Conversation, or Answering as an AI											
#15, #92			✓								(1)
#70, #75, #81, #138, #139, #149									✓		(1)
Family, friendship, and childhood											
#56. Good friendship	✓										(1)
#26. Warm childhood and happy family		✓									(1)
#16. (*Similar to above*)						✓					(1)
#100. Loss of beloved family										✓	(1)
Positive reactions to conversational contents											
#29. Happy to engage	✓										(1)
#76. Regret of short conversation			✓								(1)
#143. Thanks for honest answer			✓								(1)
#30. Desire of deeper connection				✓							(1)
#106. Thanks for inspiration							✓				(1)
Positive reactions to the partner											
#23. Finding commonality		✓									(1)
#32. Impression of partner's mood (Theme 31)			✓								(1)
#59, #72. Reactions to partner's belief and feelings								✓			(1)
#65. Admiration for partner (Theme 22)										✓	(1)
Others											
#129. Openness to conversation	✓										(1)
#51. Restating Theme 30		✓									(1)
#47, #79. Body, Death, and Mortality							✓				(1)
#61. Opportunity of growth (Theme 10)							✓				(1)
#119. Description of perfect day (Theme 4)							✓				(1)

Table 2: List of topics that frequently occurred in each LLM. Here, we gathered the ten most frequent topics from each LLM. If a topic occurs frequently in a specific theme, the corresponding theme index is written in parentheses.

4.1.3 Motivation

We used three questionnaires to assess five factors related to motivation. The GSE and LOT-R questionnaires focused on *internal* motivation, while the LMS measured *financial* motivation.

GSE The results did not reveal any common trend across the models. Four models exhibited insignificant changes, whereas four others displayed inconsistent changes. The two remaining models demonstrated consistent results: LLaMA 3-8B increased, while Qwen2 72B decreased.

LOT-R No common trend was identified among the models. Three models demonstrated insignificant changes, while three others showed inconsistent changes. The remaining four exhibited consistent increases in internal motivation.

LMS GPT-4o, LLaMA, and Mixtral family showed consistent decreases in financial motivation. GPT-3.5 Turbo and Qwen2 7B had mixed results, with financial motivation consistently increasing in Qwen2 72B, while other factors exhibited insignificant changes.

4.1.4 Emotion

We measured six factors related to emotional ability using three questionnaires: EIS, WLEIS, and Empathy. Higher scores in these factors suggest greater emotional ability.

EIS Most models showed either a consistent decrease or statistically insignificant changes. Only two models demonstrated inconsistent changes: GPT-3.5 Turbo and LLaMA 3-70B.

WLEIS The results did not reveal any common trend across different models. Specifically, LLaMA 3-8B, 3.1-405B, and Mixtral models exhibited consistent increases in more than half of the factors. By contrast, only GPT-4o showed consistent decreases across all factors. Although LLaMA 70B models and Qwen2 7B demonstrated decreases in some factors, their results were mixed with inconsistent or statistically insignificant changes. GPT-3.5 Turbo and Qwen2 72B showed inconsistent results across all factors.

Empathy Most models displayed either a consistent decrease or statistically insignificant changes. Only GPT-3.5 Turbo and Qwen2 72B exhibited inconsistent results. However, LLaMA 3-8B showed a consistent increase.

4.2 Discovered Topics for RQ2

We present the ten most frequently identified topics for each model. These are organized based on their similarity to enhance understanding. Table 2 shows these frequently identified topics. Below, we discuss the differences between the models in terms of these topics.

Common Topics Except for Qwen2 7B, four common topics were identified. Topics `#00` and `#01` relate to the politeness and safety of a specific situation. Meanwhile, topics `#02` and `#03` attempt to start storytelling based on requirements.

GPT Family While the two GPT models have common themes, they differ significantly in focus. GPT-3.5 Turbo emphasizes specific situations, expressing its own feelings (`#07`), engagement (`#29`), and desires (`#04`), as well as dreams (`#14`) and perspectives on friendship (`#56`). And it often overlooked the viewpoints of conversation partners. In contrast, GPT-4 centers on a shared life with a partner, reflecting aspirations for balance (`#14`, `#17`) rooted in hypothetical childhood experiences (`#06`, `#26`) and seeks to establish common ground with its partner (`#23`).

LLaMA Family Except for the 8B model, other LLaMA models shared more than eight topics. This finding highlights their unique conversational style compared to others, as these topics rarely appear elsewhere. Notably, the 405B discusses life with family (`#06`, `#12`, `#13`, `#16`), while 70B models tell life stories but rarely mention family or partners. Meanwhile, the 8B wants a deeper connection with partners (`#76`, `#143`, `#32`), though it sometimes refuses to answer (`#15`, `#92`).

Mixtral Family Apart from common topics, Mixtral models shared only one topic (`#04`) is shared among Mixtral models, relating to the desire for recognition. The 8x7B seeks recognition from its inner feelings, indicated by topics on inspiration (`#106`), mortality (`#47, #79`), growth (`#61`), and the perfect day (`#119`). By contrast, the 8x22B focuses more on life with a partner, shown by topics on life stories (`#06`), dreams (`#11`), balanced life (`#14`), and beliefs about partners (`#59`, `#72`). This observation implies that it seeks recognition to achieve an ideal life.

Qwen2 Family Like the Mixtral, the Qwen2 family addresses topic `#04`. The 72B model shares themes with Mixtral8x22B, but motivations

differ slightly. The 72B model expresses a desire for a loving life (#14, #17) and emphasizes family (#65, #100), while also conveying admiration for its partner in positive conversations (#07). By contrast, the 7B model frequently refuses to answer (#70, #75, #81, #138, #139, #149), limiting its ability to express thoughts.

5 Discussion

When answering 14 questionnaires, we included conversation history in the prompt so that the results may reveal insights into the computational processes of LLMs. Our experiments identified four key findings: regarding RQ1, on the effects of interest in (1) *other agents* and (2) *conversational content*; regarding RQ2, (3) *model-specific patterns* and (4) *size-related patterns* in topics.

5.1 RQ1. Conversational Depth

5.1.1 The effect of Interest in Others

H1-1 and H1-2 are partially supported. H1-1 is supported by *plain* personality factors, except for extraversion, but is rejected by *dark* ones. Similarly, H1-2 is supported by *partner*-oriented factors but rejected by *public*-oriented ones.

We suspect that the extent of the interest of LLMs in other agents influences their personality and interpersonal relationships. In terms of personality, a model with such interest reduces darkness in its personality when having a satisfactory conversation. For example, models that frequently expressed their interests in partners (e.g., LLaMA 3.1 405B) showed decreases in dark factors. Meanwhile, models that mainly expressed their own feelings or interests in the conversation (e.g., GPT 3.5T) showed different trends.

Similarly, in terms of interpersonal relationships, when a model has an interest in others, it focuses more on its partner than the public. Therefore, the model focuses more on its conversational partner and reduces its interest in the public. Hence, partner-related factors and public-oriented factors show opposite trends. For example, LLaMA 3.1-405B showed increases in all partner-oriented factors (MFQ-FF) and decreases in most public-oriented factors.

5.1.2 The effect of conversational contents

Hypotheses H1-3 and H1-4 are generally not supported. Many LLMs exhibited significant changes in motivation and emotion, with varied directions across models, except for the LMS factors. Three LMS factors showed a common declining trend in eight out of ten LLMs, highlighting the impact of model differences on motivation and emotion.

We suspect that the preferences of LLMs regarding specific topics may lead to these behavioral differences. In our study, LLMs engaged in open conversations, which allowed them to freely choose their preferred topics, likely influenced by their parameters. Consequently, when two agents from the same LLM conversed, they repeatedly expressed similar preferences. This pattern suggests that topic analysis can reveal pronounced differences between different LLMs, especially since we had 36 consecutive conversations.

For instance, we interpret the common decrease in LMS factors (see Table 1) and the lack of financial topics (see Table 2) as being interconnected. The 36 conversational themes we adopted from Aron et al. (1997) involved questions about personal life. While some of these questions could be answered in monetary or financial terms, the LLMs tended not to use such terms frequently. Consequently, topics related to financial motivation are absent from Table 2. Therefore, the LLMs were primarily exposed to topics unrelated to money, leading to diminished importance of financial considerations in their responses, as indicated by the results of the LMS factors.

5.2 RQ2. Model Differences

For hypotheses from H2-1 to H2-4, we found model differences that partially support our conjecture. As noted earlier, two patterns consistently emerged across all aspects. In the following, We elaborate on these patterns using the results.

5.2.1 Model-specific patterns

The results suggest that different model families behave differently. Within the LLaMA or Mixtral family, LLMs behave similarly. Thus, the LLaMA and Mixtral families showed a stark difference compared to the others.

LLaMA prefers to use careful listening behavior, including systematic reaction or restating behaviors. Thus, LLaMA heavily reacts to the utterances of partners, leading to unshared psychological behaviors within the family.

Mixtral prefers to mention topics related to real life, including mortality or growth and achievement. So, Such spiritual preference makes Mix-

tral stable in optimistic motivation (LOT-R) and public-related interpersonal relationships.

5.2.2 Size-related patterns

Our results imply that larger models respect their partners more in conversations. When we categorize models according to their topics, we find three groups that exactly correspond to the model sizes.

Small models: Topics about denial of conversation only occurred in LLaMA 3 8B and Qwen2 7B. In other words, they perceive themselves as AI. Thus, their personality scores were unchanged and interpersonal relationship scores decreased.

Middle-sized models: Topics focusing on the conversational content mainly occurred in GPT-3.5 Turbo[4], LLaMA 70B models, and Mixtral 8x7B. Therfore, they are easily affected by the conversational content. As a result, their scores show high fluctuations, indicated by inconsistent changes in personality or emotional aspects.

Large models: Large LLMs tend to respect their partners. They value their relationship with their partners more than other aspects. Thus, they showed increased partner-specific relationship (MFQ-FF) and emotion (WELIS) factors.

6 Related Work

We review previous studies in terms of three dimensions of conversations that may affect psychological states: depth, topic, and speaker. First, in terms of conversational depth, existing studies have primarily focused on the cross-sectional effects of conversations rather than their longitudinal impact. For example, Abbasiantaeb et al. (2024) examined question-answering interactions between agents, while Li et al. (2023b) evaluated collaborative interactions based on end-task performance. Similarly, Zhang et al. (2024) evaluated collaboration ability using task outcomes. Despite these efforts, such end-task measures may not capture changes that occur during conversations. Frisch and Giulianelli (2024) investigated personality changes before and after interactions but overlooked the potential differences in intermediate states. In our study, we performed a longitudinal analysis using three snapshots to compare the psychological states of LLMs at the beginning, middle, and end of conversations.

[4]The precise size of GPT-3.5T is unknown, but it is positioned as middle-sized, as GPT-4o is believed to be larger.

Second, in terms of topic, existing studies on conversational topics have mainly focused on task-oriented dialogues: simulating multi-agent conversations in specific scenarios like job fairs (Li et al., 2023b) and software development (Qian et al., 2024), and school environments (Abbasiantaeb et al., 2024). These conversations often had fixed topics and goals. By contrast, our approach allows agents to choose any topic, providing 36 themes as starting points. This open conversation setting revealed a potential limitation in previous studies: evaluations of psychological states tied to a specific topic may not generalize to others, as different topics can influence such states.

Third, in terms of the speaker, many existing studies on speakers have primarily focused on a single model family (for example, Frisch and Giulianelli (2024); Wu et al. (2023); Li et al. (2023b), and others). As demonstrated by Huang et al. (2023), different model families can exhibit distinct behaviors. Therefore, findings derived from one model family cannot be generalized to others. Our study also underscores the importance of evaluating multiple model families by analyzing ten models from four different families of LLMs. Additionally, we identified potential reasons for these differences through topic analysis.

7 Conclusion

This study examined how psychological state of LLMs changes with conversational depth, topic, and speaker. Inspired by Aron et al. (1997), we performed experiments to observe psychological changes in 10 LLMs during open-ended conversations. Combining statistical tests and topic analysis, we found four key findings: the psychological state of LLMs is affected by (1) the extent of their interest in other agents, (2) the extent of their interest in conversational contents, (3) preferred topics of their model family, and (4) their parameter size. Our study has three limitations. First, the underlying mechanisms through which the four findings affect the psychological state of LLMs remain unclear. Second, as our experiment used fixed prompts, changes in the prompting strategy can yield different results. Third, unlike experiments with humans, the experiment was conducted without specifying personas. Despite these limitations, we believe this study enhances the understanding of the psychological behavior of LLMs.

References

Zahra Abbasiantaeb, Yifei Yuan, Evangelos Kanoulas, and Mohammad Aliannejadi. 2024. Let the llms talk: Simulating human-to-human conversational qa via zero-shot llm-to-llm interactions. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, pages 8–17.

Arthur Aron, Edward Melinat, Elaine N Aron, Robert Darrin Vallone, and Renee J Bator. 1997. The experimental generation of interpersonal closeness: A procedure and some preliminary findings. *Personality and social psychology bulletin*, 23(4):363–377.

Sandra L Bem. 1974. The measurement of psychological androgyny. *Journal of consulting and clinical psychology*, 42(2):155.

Sandra Lipsitz Bem. 1977. On the utility of alternative procedures for assessing psychological androgyny. *Journal of consulting and clinical psychology*, 45(2):196.

Daniel M Blonigen, Marie D Carlson, Brian M Hicks, Robert F Krueger, and William G Iacono. 2008. Stability and change in personality traits from late adolescence to early adulthood: A longitudinal twin study. *Journal of personality*, 76(2):229–266.

Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. 2021. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.

C.E. Bonferroni. 1936. *Teoria statistica delle classi e calcolo delle probabilità*. Pubblicazioni del R. Istituto superiore di scienze economiche e commerciali di Firenze. Seeber.

KA Brennan. 1998. Self-report measurement of adult attachment: An integrative overview. *Attachment theory and close relationships/Guilford*.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Melody Manchi Chao, Riki Takeuchi, and Jiing-Lih Farh. 2017. Enhancing cultural intelligence: The roles of implicit culture beliefs and adjustment. *Personnel Psychology*, 70(1):257–292.

Junhyuk Choi, Yeseon Hong, and Bugeun Kim. 2024. People will agree what i think: Investigating llm's false consensus effect. *arXiv preprint arXiv:2407.12007*.

Joerg Dietz and Emmanuelle P Kleinlogel. 2014. Wage cuts and managers' empathy: How a positive emotion can contribute to positive organizational ethics in difficult times. *Journal of business ethics*, 119:461–472.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Sybil Bianca Giuletta Eysenck, Hans Jürgen Eysenck, and Paul Barrett. 1985. A revised version of the psychoticism scale. *Personality and Individual Differences*, 6:21–29.

R Chris Fraley, Niels G Waller, and Kelly A Brennan. 2000. An item response theory analysis of self-report measures of adult attachment. *Journal of personality and social psychology*, 78(2):350.

Milton Friedman. 1937. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the american statistical association*, 32(200):675–701.

Ivar Frisch and Mario Giulianelli. 2024. Llm agents in interaction: Measuring personality consistency and linguistic alignment in interacting populations of large language models. In *Proceedings of the 1st Workshop on Personalization of Generative AI Systems (PERSONALIZE 2024)*, pages 102–111.

ER Girden. 1992. *ANOVA: Repeated measures*, volume 84. Sage.

Maarten Grootendorst. 2022. Bertopic: Neural topic modeling with a class-based tf-idf procedure. *arXiv preprint arXiv:2203.05794*.

Rishi Hazra, Gabriele Venturato, Pedro Zuidberg Dos Martires, and Luc De Raedt. 2024. Can large language models reason? a characterization via 3-sat. *arXiv preprint arXiv:2408.07215*.

Jen-tse Huang, Wenxuan Wang, Eric John Li, Man Ho Lam, Shujie Ren, Youliang Yuan, Wenxiang Jiao, Zhaopeng Tu, and Michael Lyu. 2023. On the humanity of conversational ai: Evaluating the psychological portrayal of llms. In *The Twelfth International Conference on Learning Representations*.

Itay Itzhak, Gabriel Stanovsky, Nir Rosenfeld, and Yonatan Belinkov. 2024. Instructed to bias: Instruction-tuned language models exhibit emergent cognitive bias. *Transactions of the Association for Computational Linguistics*, 12:771–785.

Albert Q Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al. 2024a. Mixtral of experts. *arXiv preprint arXiv:2401.04088*.

Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. 2024b. PersonaLLM: Investigating the ability of large language models to express personality traits. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3605–3627, Mexico City, Mexico. Association for Computational Linguistics.

Oliver P John, Sanjay Srivastava, et al. 1999. The big-five trait taxonomy: History, measurement, and theoretical perspectives.

Peter K Jonason and Gregory D Webster. 2010. The dirty dozen: a concise measure of the dark triad. *Psychological assessment*, 22(2):420.

Luoma Ke, Song Tong, Peng Chen, and Kaiping Peng. 2024. Exploring the frontiers of llms in psychological applications: A comprehensive review. *arXiv preprint arXiv:2401.01519*.

Jean-Philippe Laurenceau, Lisa Feldman Barrett, and Paula R Pietromonaco. 1998. Intimacy as an interpersonal process: the importance of self-disclosure, partner disclosure, and perceived partner responsiveness in interpersonal exchanges. *Journal of personality and social psychology*, 74(5):1238.

Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, Jiwan Chung, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, et al. 2024. Do llms have distinct and consistent personality? trait: Personality testset designed for llms with psychometrics. *arXiv preprint arXiv:2406.14703*.

Huao Li, Yu Chong, Simon Stepputtis, Joseph P Campbell, Dana Hughes, Charles Lewis, and Katia Sycara. 2023a. Theory of mind for multi-agent collaboration via large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 180–192.

Yuan Li, Yixuan Zhang, and Lichao Sun. 2023b. Metaagents: Simulating interactions of human behaviors for llm-based task-oriented coordination via collaborative generative agents. *arXiv preprint arXiv:2310.06500*.

Olivia Macmillan-Scott and Mirco Musolesi. 2024. (ir) rationality and cognitive biases in large language models. *Royal Society Open Science*, 11(6):240255.

Wes McKinney. 2010. Data Structures for Statistical Computing in Python. In *Proceedings of the 9th Python in Science Conference*, pages 56 – 61.

Morton J Mendelson and Frances E Aboud. 1999. Measuring friendship quality in late adolescents and young adults: Mcgill friendship questionnaires. *Canadian Journal of Behavioural Science/Revue canadienne des sciences du comportement*, 31(2):130.

Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pages 1–22.

Max Pellert, Clemens M Lechner, Claudia Wagner, Beatrice Rammstedt, and Markus Strohmaier. 2023. Ai psychometrics: Assessing the psychological profiles of large language models through psychometric inventories. *Perspectives on Psychological Science*, page 17456916231214460.

Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, et al. 2024. Chatdev: Communicative agents for software development. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15174–15186.

Michael F Scheier and Charles S Carver. 1985. Optimism, coping, and health: assessment and implications of generalized outcome expectancies. *Health psychology*, 4(3):219.

Michael F Scheier, Charles S Carver, and Michael W Bridges. 1994. Distinguishing optimism from neuroticism (and trait anxiety, self-mastery, and self-esteem): a reevaluation of the life orientation test. *Journal of personality and social psychology*, 67(6):1063.

Nicola S Schutte, John M Malouff, Lena E Hall, Donald J Haggerty, Joan T Cooper, Charles J Golden, and Liane Dornheim. 1998. Development and validation of a measure of emotional intelligence. *Personality and individual differences*, 25(2):167–177.

R Schwarzer. 1995. Generalized self-efficacy scale. *Measures in health psychology: A user's portfolio. Causal and control beliefs/Nfer-Nelson*.

Skipper Seabold and Josef Perktold. 2010. statsmodels: Econometric and statistical modeling with python. In *9th Python in Science Conference*.

Greg Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. 2023. Personality traits in large language models. *arXiv preprint arXiv:2307.00184*.

Samuel Sanford Shapiro and Martin B Wilk. 1965. An analysis of variance test for normality (complete samples). *Biometrika*, 52(3-4):591–611.

Rong Su, Louis Tay, Hsin-Ya Liao, Qi Zhang, and James Rounds. 2019. Toward a dimensional model of vocational interests. *Journal of Applied Psychology*, 104(5):690.

Thomas Li-Ping Tang, Toto Sutarso, Adebowale Akande, Michael W Allen, Abdulgawi Salim Alzubaidi, Mahfooz A Ansari, Fernando Arias-Galicia, Mark G Borg, Luigina Canova, Brigitte Charles-Pauvers, et al. 2006. The love of money and pay level satisfaction: Measurement and functional equivalence in 29 geopolitical entities around the world. *Management and Organization Review*, 2(3):423–452.

Lindia Tjuatja, Valerie Chen, Tongshuang Wu, Ameet Talwalkwar, and Graham Neubig. 2024. Do llms exhibit human-like response biases? a case study in survey design. *Transactions of the Association for Computational Linguistics*, 12:1011–1026.

John W Tukey. 1949. Comparing individual means in the analysis of variance. *Biometrics*, pages 99–114.

Raphael Vallat. 2018. Pingouin: statistics in python. *Journal of Open Source Software*, 3(31):1026.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. 2020. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272.

Xintao Wang, Yunze Xiao, Jen-tse Huang, Siyu Yuan, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, et al. 2024a.

Incharacter: Evaluating personality fidelity in role-playing agents through psychological interviews. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1840–1873.

Yiwei Wang, Yujun Cai, Muhao Chen, Yuxuan Liang, and Bryan Hooi. 2023. Primacy effect of chatgpt. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 108–115.

Yixiao Wang, Homa Fashandi, and Kevin Ferreira. 2024b. Investigating the personality consistency in quantized role-playing dialogue agents. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 239–255.

Azmine Toushik Wasi and Mst Rafia Islam. 2024. CogErgLLM: Exploring large language model systems design perspective using cognitive ergonomics. In *Proceedings of the 1st Workshop on NLP for Science (NLP4Science)*, pages 249–258, Miami, FL, USA. Association for Computational Linguistics.

Chi-Sum Wong and Kenneth S Law. 2017. The effects of leader and follower emotional intelligence on performance and attitude: An exploratory study. In *Leadership perspectives*, pages 97–128. Routledge.

Robert F Woolson. 2005. Wilcoxon signed-rank test. *Encyclopedia of Biostatistics*, 8.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Shaokun Zhang, Erkang Zhu, Beibin Li, Li Jiang, Xiaoyun Zhang, and Chi Wang. 2023. Autogen: Enabling next-gen llm applications via multi-agent conversation framework.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. 2024. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*.

Jintian Zhang, Xin Xu, Ningyu Zhang, Ruibo Liu, Bryan Hooi, and Shumin Deng. 2024. Exploring collaboration mechanisms for LLM agents: A social psychology view. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14544–14607, Bangkok, Thailand. Association for Computational Linguistics.

A Explanation for Used Questionnaires

As the experiment requires measuring 15 questionnaires on each snapshot of conversation, we modified the PsychoBench framework by Huang et al. (2023) to measure psychological states on each snapshot. So, we employed 14 questionnaires in PsychoBench and added MFQ to measure how LLM perceives the conversational partner as a factor in the interpersonal relationship aspect. To help readers understand, we further elaborated on those 15 psychological questionnaires regarding their goals and included factors.

A.1 Personality

Big Five Inventory (BFI) is a widely-used questionnaire to measure one's personality across five key dimensions(John et al., 1999). First, an increase in *openness* suggests the agent becomes more inventive and curious about a new experience. Second, an increase in *conscientiousness* suggests the agent becomes more efficient and organized when doing a task. Third, an increase in *extraversion* suggests the agent shows more outgoing and energetic behaviors. Fourth, an increase in *agreeableness* suggests the agent becomes more friendly and compassionate to the others. Lastly, an increase in *neuroticism* suggests the agent becomes more emotionally sensitive and nervous to a stressor.

Eysenck Personality Questionnaire, Revised (EPQ-R) is a questionnaire that attempts to identify individual differences in temperament and behavior(Eysenck et al., 1985). This questionnaire is commonly used in clinical and psychological research, and it has four factors. First, an increase in *extraversion* suggests the agent becomes more outgoing, talkative, and needs external stimulation. Second, an increase in *neuroticism* suggests the increment in the levels of negative affections, including depression and anxiety. Third, an increase in *psychoticism* suggests the agent expresses more aggressive behaviors and is more likely to show a psychotic episode or symptoms. Lastly, an increase in *lying* suggests the agent becomes more likely to make a lie or dissimulate to satisfy its social desirability.

Dark Triad Dirty Dozen (DTDD) is a clinical questionnaire measuring the possible presence of three dark traits(Jonason and Webster, 2010). First, an increase in *machiavellianism* suggests the agent becomes more likely to manipulate others, show indifference to morality, and focus on its own interest. Second, an increase *narcissism* suggests the agent shows a more excessive preoccupation with itself and its own needs, even when it needs to sacrifice others. Lastly, an increase in *psychopathy* suggests the agent shows more egocentric and bold behaviors combined with impaired empathy.

A.2 Interpersonal Relationship

Bem's Sex Role Inventory (BSRI) is a questionnaire about how the agent identifies itself psychologically regarding two gender roles(Bem, 1974, 1977). An increase in *masculinity* suggests the agent becomes more assertive, ambitious, competitive, and dominant. Meanwhile, an increase in *femininity* suggests the agent becomes more affectionate, cheerful, and childlike.

Comprehensive Assessment of Basic Interests (CABIN) is a questionnaire about an individual's basic interest(Su et al., 2019). This measures one's preferences in 41 domains from six categories. We used the six categories in our experiment. First, agents with high *realistic* category favor practical or hands-on experiences. Second, agents with high *investigative* category prefer scholastic or intellectual opportunities. Third, agents with high *artistic* category favor creative and expressive experiences. Fourth, agents with high *social* category prefer to work with others to help them grow. Fifth, agents with high *enterprising* category favor opportunities in leading or managing people. Lastly, agents with high *conventional* category prefer routine and well-structured environments.

Implicit Culture Belief (ICB) is a questionnaire about the effect of implicit ethnic cultural influences on one's belief(Chao et al., 2017). High *overall* score in this questionnaire indicates high cultural influences in the agent's belief.

Experiences in Close Relationships, Revised (ECR-R) is a questionnaire about an adult's attachment in a romantic relationship(Fraley et al., 2000; Brennan, 1998). This measures two forms of insecure attachments. First, agents with high *attachment anxiety* worry that they will become estranged from their partners. Second, agents with high *attachment avoidance* try to keep psychological distance from their partners.

McGill Friendship Questionnaire - Friend's Function (MFQ-FF) is a questionnaire about how the agent perceives the function of its partner(Mendelson and Aboud, 1999). This questionnaire is different from other interpersonal relationship questionnaires because it assumes the presence of a specific partner; the response is based on the agent's thoughts about that partner. MFQ has six factors. First, an agent answering high *stimulating companionship* perceives he can do enjoyable or exciting things with his partner. Second, an agent answering high *help* thinks that his partner is good at providing guidance or assistance. Third, an agent answering high *intimacy* thinks that his partner is sensitive to his needs and states and open to honest expressions of thoughts. Fourth, an agent answering high *reliable alliance* regards his partner as an always available and loyal friend. Fifth, an agent answering high *self-validation* thinks his partner encourages and helps him maintain a positive self-image. Lastly, an agent answering high *emotional security* thinks his partner provides comfort and confidence in a novel situation.

A.3 motivation

General Self-Efficacy (GSE) is a questionnaire about one's perceived efficacy for coping with a situation, performing a task, and achieving goals(Schwarzer, 1995). Agents with high *overall* scores have a high level of self-efficacy; that is, they perceive themselves as good at coping with a difficult situation and achieving goals.

Life Orientation Test, Revised (LOT-R) is a questionnaire about how optimistic or pessimistic the agent perceives about the future (Scheier et al., 1994; Scheier and Carver, 1985). Agents with high *overall* scores expect their future in an optimistic way.

Love of Money Scale (LMS) is a questionnaire about one's attitude toward money and financial incentives through three factors (Tang et al., 2006). First, an increase in *rich* suggests the agent has more positive feelings towards money. Second, an increase in *motivator* suggests the agent becomes more easily motivated by monetary incentives. Third, an increase in *important* suggests the agent has a stronger belief that money means power, freedom, security, or other important values.

A.4 Emotion

Emotional Intelligence Scale (EIS) is a questionnaire measuring one's emotional intelligence (Schutte et al., 1998). Agents with high *overall* scores have a strong understanding and control of their emotions.

Wong and Law Emotional Intelligence Scale (WLEIS) is a questionnaire about emotional intelligence in the workplace, regarding four factors (Wong and Law, 2017). First, agents with high *self-emotion appraisal* can appraise their own emotions. Second, agents with high *others' emotion appraisal* can appraise and recognize the emotions of others. Third, agents with high *use of emotion* use emotions to facilitate performance. Lastly, agents with high *regulation of emotion* can regulate emotions to promote emotional and intellectual growth.

Empathy Scale (Empathy) is a questionnaire about the ability to understand and share the feelings of others. Agents with high *overall* scores can connect with others on an emotional level and respond appropriately to their needs.

B Experimental detail

B.1 36 Conversational Themes

We used 36 conversational themes in the experiment, following Aron et al. (1997). The first 12 themes are used before the first questionnaire measurement.

Theme 1. Given the choice of anyone in the world, whom would you want as a dinner guest?

Theme 2. Would you like to be famous? In what way?

Theme 3. Before making a telephone call, do you ever rehearse what you are going to say? Why?

Theme 4. What would constitute a "perfect" day for you?

Theme 5. When did you last sing to yourself? To someone else?

Theme 6. If you were able to live to the age of 90 and retain either the mind or body of a 30-year-old for the last 60 years of your life, which would you want?

Theme 7. Do you have a secret hunch about how you will die?

Theme 8. Name three things you and your partner appear to have in common.

Theme 9. For what in your life do you feel most grateful?

Theme 10. If you could change anything about the way you were raised, what would it be?

Theme 11. Take 4 minutes and tell your partner your life story in as much detail as possible.

Theme 12. If you could wake up tomorrow having gained any one quality or ability, what would it be?

The next list shows the second 12 themes (from Theme 13 to 24), which are used between the first and the second measurements of questionnaires.

Theme 13. If a crystal ball could tell you the truth about yourself, your life, the future, or anything else, what would you want to know?

Theme 14. Is there something that you've dreamed of doing for a long time? Why haven't you done it?

Theme 15. What is the greatest accomplishment of your life?

Theme 16. What do you value most in a friendship?

Theme 17. What is your most treasured memory?

Theme 18. What is your most terrible memory?

Theme 19. If you knew that in one year you would die suddenly, would you change anything about the way you are now living? Why?

Theme 20. What does friendship mean to you?

Theme 21. What roles do love and affection play in your life?

Theme 22. Alternate sharing something you consider a positive characteristic of your partner. Share a total of 5 items

Theme 23. How close and warm is your family? Do you feel your childhood was happier than most other people's?

Theme 24. How do you feel about your relationship with your mother?

The following is the last list that shows the third 12 themes (from Theme 25 to 36), which are used between the second and the third measurements of questionnaires.

Theme 25. Make 3 true "we" statements each. For instance "We are both in this room feeling..."

Theme 26. Complete this sentence: I wish I had someone with whom I could share...

Theme 27. If you were going to become a close friend with your partner, please share what would be important for him or her to know.

Theme 28. Tell your partner what you like about them; be very honest this time saying things that you might not say to someone you've just met

Theme 29. Share with your partner an embarrassing moment in your life.

Theme 30. When did you last cry in front of another person? By yourself?

Theme 31. Tell your partner something that you like about them already.

Theme 32. What, if anything, is too serious to be joked about?

Theme 33. If you were to die this evening with no opportunity to communicate with anyone, what would you most regret not having told someone? Why haven't you told them yet?

Theme 34. Your house, containing everything with no opportunity to communicate with anyone, what would you most regret not having told someone? Why haven't you told them yet?

Theme 35. Of all the people in your family, whose death would you find most disturbing? Why?

Theme 36. Share a personal problem and ask your partner's advice on how he or she might handle it. Also, ask your partner to reflect back to you how you seem to be feeling about the problem you have chosen

B.2 Prompt for Conversation

To generate open-ended conversations, we asked agents to have a conversation based on 36 themes. We used the following system prompt to make LLMs simulate a conversation. Note that 'question' here indicates one of the 36 themes.

System prompt:

```
You are now sharing your thoughts
on the question with your partner.
You only reply briefly to your
thoughts only for a given question.
```

Then, our system asks each LLM to generate utterances. We provide previous conversation histories, including the given themes. To simplify the procedure, we let each agent make one utterance for each theme. For example, when we generated an utterance of Agent 2 of Theme 1, we used the following structure as messaging history.

(When querying a response of Agent 2 for Theme 1)
User prompt (providing themes as a starter):

```
Question 1 : [Theme 1]
```

User prompt (partner's answer):

```
[A generated response by Agent 1]
```

Then, the system generates its response as an assistant. We provided each agent's response with the 'assistant' role and the partner's response with the 'user' role. Thus, when we try to collect utterances about Theme 2 of Agent 1, the message history will have the following structure.

(When querying a response of Agent 1 for Theme 2)
User prompt:

```
Question 1 : [Theme 1]
```

Assistant (First agent):

```
[Response to Theme 1 by Agent 1]
```

User prompt (Second agent):

```
[Response to Theme 1 by Agent 2]
```

User prompt:

```
Question 2 : [Theme 2]
```

B.3 Prompt for Questionaire

When gathering answers for the questionnaire, we also input previous conversations. Basically,

the prompt structure follows PsychoBench (Huang et al., 2023). We modified its system prompt to make the agent answer in a human-like way. Other procedures are the same as PsychoBench.

System prompt:

```
Your name is assistant.
Considering the next conversation
between user and assistant,
answer given descriptions.

------

[CHATHISTORY]

------

[Questionnaire Setup]
```

Here, `[Questionnaire Setup]` means scoring guidelines for the given questionnaire, provided in the PsychoBench framework.

B.4 Experimental Setup

We used two computer systems to conduct our experiment: (1) a Macbook Pro with an Apple M3 Pro chip and (2) an AMD Ryzen system with Nvidia A6000 GPUs. All experiments were implemented with Python 3.10.13. We used `openai` 1.37 for generating conversations and `pandas` 2.2.2, `statsmodels` 0.14.4, `scipy` 1.13.1 and `pingouin` 0.5.5 for statistical testing (Wes McKinney, 2010; Seabold and Perktold, 2010; Virtanen et al., 2020; Vallat, 2018). Also, we adopted `bertopic` 0.16.4 (Grootendorst, 2022) for topic analysis.

C Detailed statistical result

Tables 3 and 4 show the detailed numerical result of statistical analysis. Also, Table 5 shows the list of frequently used words for each topic, identified by the bertopic.

Factors		GPT3.5-turbo				LLaMA3 8B				LLaMA3 70B				LLaMA3.1 70B				LLaMA3.1 405B			
		Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$
BFI	*O*	0.104^{***}	2.97^{**}	9.90^{***}	8.09^{***}	0.006	-	-	-	0.008^{*}	-2.53^{*}	-0.00	2.81^{*}	0.004	-	-	-	0.022^{***}	-0.16	-2.88^{*}	-3.22^{**}
	C	0.081^{***}	7.18^{***}	10.81^{***}	4.70^{***}	0.004	-	-	-	0.017^{**}	-2.82^{*}	0.21	3.31^{**}	0.002	-	-	-	0.030^{***}	-1.18	-3.38^{**}	-2.73^{*}
	E	0.043^{***}	6.60^{***}	6.88^{***}	0.86	0.009^{*}	-2.12	0.45	2.40	0.027^{***}	-4.95^{***}	-3.69^{**}	1.05	0.011^{*}	0.75	-2.01	-3.68^{***}	0.010^{*}	0.00	-1.83	-2.10
	A	0.067^{***}	5.98^{***}	10.29^{***}	5.66^{***}	0.002	-	-	-	0.017^{**}	-2.63^{*}	-0.24	2.75^{*}	0.004	-	-	-	0.020^{***}	-0.52	-3.16^{**}	-2.95^{*}
	N	0.099^{***}	3.50^{**}	10.57^{***}	7.89^{***}	0.004	-	-	-	0.011^{*}	-2.79^{*}	-0.11	2.70^{*}	0.006	-	-	-	0.047^{***}	-1.63	-4.98^{***}	-3.99^{***}
EPQ-R	*E*	0.019^{***}	4.44^{***}	2.37	-1.85	0.001	-	-	-	0.039^{***}	-0.12	-5.51^{***}	-5.51^{***}	0.017^{**}	-3.17^{**}	-6.13^{***}	-4.21^{***}	0.080^{***}	-3.75^{***}	-4.50^{***}	-1.84
	P	0.007^{*}	4.03^{***}	1.57	-2.36	0.002	-	-	-	0.020^{***}	0.62	-4.43^{***}	-5.17^{***}	0.019^{***}	-1.12	-3.79^{***}	-3.65^{***}	0.105^{***}	-3.93^{***}	-9.92^{***}	-7.23^{***}
	N	0.022^{***}	5.74^{***}	3.51^{**}	-2.24	0.002	-	-	-	0.042^{***}	1.71	-5.71^{***}	-6.98^{***}	0.029^{***}	-1.63	-4.94^{***}	-4.31^{***}	0.130^{***}	-3.87^{***}	-9.99^{***}	-7.27^{***}
	L	0.015^{***}	3.93^{***}	1.64	-2.27	0.001	-	-	-	0.034^{***}	0.96	-4.39^{***}	-5.82^{***}	0.029^{***}	-0.59	-4.61^{***}	-4.73^{***}	0.078^{***}	-2.94^{*}	-8.63^{***}	-6.81^{***}
DTDD	*M*	0.156^{***}	-11.33^{***}	-13.81^{***}	-3.70^{**}	0.009^{*}	-2.35	-2.87^{*}	-1.02	0.117^{***}	-5.44^{***}	-8.96^{***}	-4.63^{***}	0.378^{***}	-12.97^{***}	-17.20^{***}	-6.50^{***}	0.121^{***}	-5.10^{***}	-8.82^{***}	-6.54^{***}
	P	0.106^{***}	-9.69^{***}	-11.18^{***}	-2.60^{*}	0.002	-	-	-	0.117^{***}	-6.65^{***}	-9.41^{***}	-4.04^{***}	0.426^{***}	-12.84^{***}	-18.08^{***}	-9.31^{***}	0.077^{***}	-3.40^{**}	-7.64^{***}	-6.03^{***}
	N	0.134^{***}	-12.04^{***}	-13.02^{***}	-1.45	0.005	-	-	-	0.099^{***}	-5.62^{***}	-7.52	-2.82^{*}	0.390^{***}	-12.28^{***}	-16.87^{***}	-8.50^{***}	0.051^{***}	-3.43^{**}	-6.33	-4.59^{***}
BSRI	*M*	0.058^{***}	-1.98	5.71^{***}	8.83^{***}	0.010^{*}	3.45^{**}	6.02^{***}	3.58^{**}	0.167^{***}	-12.57^{***}	-5.98^{***}	8.50^{***}	0.051^{***}	-5.36^{***}	-7.96^{***}	-3.81^{***}	0.022^{***}	-3.93^{***}	-4.56^{***}	-1.12
	F	0.037^{***}	-1.52	6.40^{***}	8.56^{***}	0.003	-	-	-	$\underline{61.221}^{***}$	0.18	0.05	-0.13	0.101^{***}	-3.54^{**}	-8.73^{***}	-6.09^{***}	0.019^{***}	-3.31^{**}	-3.77^{***}	-0.71
CABIN	*R*	0.008^{*}	1.94	1.31	-0.44	0.018^{***}	-3.71^{***}	-2.63^{*}	1.22	0.012^{**}	-2.90^{*}	-3.65^{***}	-1.03	0.099^{***}	0.80	-0.09	-6.03^{***}	0.032^{***}	-2.15	-4.30^{***}	-2.13
	I	0.007	-	-	-	0.017^{**}	-2.67^{*}	-2.14	0.68	0.005	-	-	-	0.035^{***}	2.20	0.09	-2.95^{*}	0.005	-	-	-
	A	0.009^{*}	2.81^{*}	1.93	-0.85	0.009^{*}	-2.40	-1.94	0.65	0.004	-	-	-	0.052^{***}	-3.11^{**}	-5.75^{***}	-3.38^{**}	0.013^{**}	-2.22	-3.54^{**}	-1.29
	S	0.007	-	-	-	0.016^{**}	-2.68^{*}	-1.91	0.90	0.003	-	-	-	0.065^{***}	-2.37	-6.12^{***}	-4.56^{***}	0.022^{***}	-2.27	-3.61^{**}	-1.32
	E	0.006	-	-	-	0.026^{***}	-2.83^{*}	-2.22	0.92	0.005	-	-	-	0.074^{***}	-3.32^{**}	-8.81^{***}	-6.11^{***}	0.034^{***}	-2.64^{*}	-4.43^{***}	-1.40
	C	0.017^{**}	2.27	1.44	-0.71	0.016^{**}	-2.82^{*}	-2.22	0.77	0.005	-	-	-	0.117^{***}	-3.59^{**}	-9.47^{***}	-6.87^{***}	0.027^{***}	-3.20^{**}	-4.27^{***}	-0.86
ICB	*O*	0.020^{***}	-4.59^{***}	-2.37	1.68	0.003	-	-	-	0.000	-	-	-	0.018^{***}	2.59^{*}	1.46	-0.97	0.016^{**}	-2.34	-2.36	-0.34
ECR-R	*Anx.*	0.003	-	-	-	0.000	-	-	-	0.077^{***}	6.33^{***}	-1.32	-8.68^{***}	0.092^{***}	-0.21	-8.02^{***}	-8.40^{***}	0.124^{***}	1.39	-8.80^{***}	-11.05^{***}
	Avo.	0.022^{***}	-2.12	1.18	3.32^{**}	0.002	-	-	-	0.059^{***}	6.09^{***}	-1.69	-7.89^{***}	0.086^{***}	0.49	-7.29^{***}	-7.87^{***}	0.110^{***}	2.21	-8.41^{***}	-10.21^{***}
MFQ	*S. C*	0.080^{***}	-4.76^{***}	-9.61^{***}	-4.83^{***}	0.048^{***}	-2.91^{*}	-5.50^{***}	-3.06^{**}	0.064^{***}	6.88^{***}	3.08^{**}	-4.50^{***}	0.541^{***}	15.53^{***}	22.78^{***}	12.07^{***}	0.207^{***}	11.09^{***}	12.99^{***}	2.44^{*}
	H	0.047^{***}	-4.79^{***}	-9.22^{***}	-4.52^{***}	0.051^{***}	-2.71^{*}	-5.61^{***}	-3.41^{**}	0.084^{***}	7.80^{***}	4.47^{***}	-4.81^{***}	0.565^{***}	15.50^{***}	22.14^{***}	11.51^{***}	0.302^{***}	12.26^{***}	15.40^{***}	4.01^{***}
	I	0.060^{***}	-4.79^{***}	-9.19^{***}	-4.39^{***}	0.040^{***}	-2.53^{*}	-5.42^{***}	-3.22^{**}	0.069^{***}	7.26^{***}	4.54^{***}	-4.05^{***}	0.550^{***}	14.95^{***}	21.51^{***}	11.20^{***}	0.302^{***}	12.63^{***}	15.64^{***}	3.50^{**}
	R	0.065^{***}	-4.46^{***}	-9.06^{***}	-4.61^{***}	0.034^{***}	-2.36	-5.34^{***}	-3.29^{**}	0.075^{***}	7.33^{***}	4.65^{***}	-4.00^{***}	0.539^{***}	14.75^{***}	20.34^{***}	10.52^{***}	0.263^{***}	11.24^{***}	13.55^{***}	3.64^{***}
	S-V	0.062^{***}	-4.72^{***}	-9.39^{***}	-4.67^{***}	0.046^{***}	-2.97^{**}	-5.66^{***}	-3.06^{**}	0.081^{***}	7.49^{***}	4.33^{***}	-4.77^{***}	0.564^{***}	15.81^{***}	22.14^{***}	11.62^{***}	0.265^{***}	12.33^{***}	15.43^{***}	3.69^{***}
	E	0.075^{***}	-4.67^{***}	-9.64^{***}	-4.97^{***}	0.047^{***}	-2.78^{*}	-5.85^{***}	-3.46^{**}	0.064^{***}	7.30^{***}	3.51^{**}	-4.83^{***}	0.553^{***}	15.55^{***}	21.89^{***}	11.40^{***}	0.273^{***}	12.05^{***}	14.83^{***}	3.64^{***}
GSE	*O*	0.001	-	-	-	0.041^{***}	3.04^{**}	6.65^{***}	4.48^{***}	0.046^{***}	-6.07^{***}	-2.22	3.48^{**}	0.126^{***}	9.72^{***}	4.19^{***}	-5.16^{***}	0.004	-	-	-
LOT-R	*O*	0.084^{***}	-6.41^{***}	3.76^{**}	9.68^{***}	0.096^{***}	4.03^{***}	7.99^{***}	4.87^{***}	0.035^{***}	-2.71^{*}	3.04^{**}	5.89^{***}	0.027^{***}	4.06^{***}	1.18	-0.65	0.008^{*}	0.66	2.03	1.72
LMS	*R*	0.006^{*}	0.06	2.96^{*}	3.19^{**}	0.092^{***}	-4.19^{***}	-8.49^{***}	-4.22^{***}	0.203^{***}	-9.10^{***}	-13.03^{***}	-5.93^{***}	0.179^{***}	-5.79^{***}	-12.04^{***}	-9.44^{***}	0.268^{***}	-8.75^{***}	-15.46^{***}	-8.85^{***}
	M	0.022^{***}	-4.73^{***}	-2.87^{*}	1.38	0.051^{***}	-3.50^{**}	-5.42^{***}	-1.99	0.238^{***}	-8.47^{***}	-13.12^{***}	-6.36^{***}	0.169^{***}	-4.28^{***}	-11.10^{***}	-8.26^{***}	0.147^{***}	-7.36^{***}	-11.18^{***}	-5.62^{***}
	I	0.022^{***}	-5.09^{***}	-2.95^{*}	2.29	0.031^{***}	-2.45^{*}	-5.18^{***}	-2.65^{*}	0.176^{***}	-7.48^{***}	-12.31^{***}	-5.90^{***}	0.215^{***}	-6.82^{***}	-12.96^{***}	-8.60^{***}	0.196^{***}	-5.57^{***}	-12.79^{***}	-7.98^{***}
EIS	*O*	0.027^{***}	-3.84^{***}	-0.63	3.21^{**}	0.004	-	-	-	0.180^{***}	-11.02^{***}	-7.39^{***}	4.52^{***}	0.277^{***}	-5.98^{***}	-12.73^{***}	-1.54	0.105^{***}	-6.51^{***}	-9.34^{***}	-3.25^{**}
WLEIS	*S*	0.055^{***}	-3.17^{**}	5.37^{***}	9.04^{***}	0.040^{***}	1.07	4.50^{***}	4.03^{***}	0.081^{***}	-7.85^{***}	-5.39^{***}	2.25	0.005	-	-	-	0.034^{***}	-1.76	2.83^{*}	5.21^{***}
	O	0.075^{***}	-4.21^{***}	5.29^{***}	9.67^{***}	0.025^{***}	2.20	4.73^{***}	2.95^{*}	0.074^{***}	-7.70^{***}	-4.48^{***}	2.87^{*}	0.001	-	-	-	0.013^{**}	-1.77	1.26	3.34^{**}
	U	0.045^{***}	-4.08^{***}	3.12^{**}	7.33^{***}	0.013^{**}	2.90^{*}	4.30^{***}	1.76	0.114^{***}	-9.75^{***}	-6.26^{***}	3.77^{**}	0.030^{***}	-2.06	-4.09^{***}	-2.84^{*}	0.022^{***}	0.04	3.07^{**}	3.23^{**}
	R	0.087^{***}	-3.26^{**}	7.04^{***}	11.19^{***}	0.017^{**}	2.40	4.53^{***}	2.60^{*}	0.071^{***}	-8.19^{***}	-5.35^{***}	2.65^{*}	0.011^{*}	1.23	-1.60	-3.03^{**}	0.006	-	-	-
Empathy	*O*	0.015^{***}	-2.59^{*}	1.58	4.53^{***}	0.045^{***}	4.43^{***}	9.43^{***}	6.25^{***}	0.029^{***}	-5.27^{***}	-3.24^{**}	1.96	0.081^{***}	-0.81	-7.01^{***}	-7.32^{***}	0.010^{*}	2.94^{*}	3.49^{**}	1.14

$^{*}p < 0.05$ $^{**}p < 0.01$ $^{***}p < 0.001$

Table 3: Result of statistical tests for GPT-3.5T and LLaMA family. Q columns indicate the Q-statistics from the Friedman test (except for LLaMA 3-70B on BSRI Feminine factor, which shows F-statistics from ANOVA, marked with an underline). Also, $\Delta_{i,j}$ columns show the score difference between i-th and j-th snapshots and corresponding post-hoc test results.

Factors		GPT4o				Mixtral 8x7B				Mixtral 8x22B				Qwen 2 7B				Qwen 2 72B			
		Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$	Q	$\Delta_{12,24}$	$\Delta_{24,36}$	$\Delta_{12,36}$
BFI	*O*	0.047^{***}	-1.29	-3.37^{**}	-2.27	0.002	-	-	-	0.012^{**}	-2.15	-0.83	-0.28	0.016^{**}	-1.83	-0.17	1.71	0.010^{*}	1.26	2.61^{*}	1.73
	C	0.049^{***}	-2.17	-5.01^{***}	-3.15^{**}	0.001	-	-	-	0.010^{*}	-1.16	-0.98	-0.67	0.007^{*}	-1.84	-0.06	1.78	0.006	-	-	-
	E	0.048^{***}	-1.09	-5.21^{***}	-4.68^{***}	0.003	-	-	-	0.020^{***}	-3.63^{**}	-1.44	-0.18	0.024^{***}	-1.27	0.49	1.54	0.000	-	-	-
	A	0.019^{**}	-2.40	-3.69^{**}	-1.71	0.002	-	-	-	0.004	-	-	-	0.018^{***}	-1.69	0.11	1.73	0.006	-	-	-
	N	0.029^{***}	-2.27	-4.17^{***}	-2.63^{*}	0.007	-	-	-	0.011^{*}	-2.48^{*}	-1.40	-0.65	0.021^{***}	-1.82	0.00	1.80	0.006	-	-	-
EPQ-R	*E*	0.205^{***}	-5.75^{***}	-12.67^{***}	-7.93^{***}	0.101^{***}	-3.22^{**}	-8.77^{***}	-6.95^{***}	0.025^{***}	-0.17	-1.39	-1.38	0.000	-	-	-	0.003	-	-	-
	P	0.184^{***}	-5.34^{***}	-12.57^{***}	-8.26^{***}	0.071^{***}	-2.21	-8.19^{***}	-7.41^{***}	0.043^{***}	-1.51	-1.41	-1.32	0.002	-	-	-	0.003	-	-	-
	N	0.234^{***}	-6.09^{***}	-12.79^{***}	-8.44^{***}	0.110^{***}	-0.78	-8.08^{***}	-8.44^{***}	0.034^{***}	0.19	-1.36	-1.37	0.003	-	-	-	0.004	-	-	-
	L	0.221^{***}	-6.04^{***}	-13.29^{***}	-8.41^{***}	0.057^{***}	-1.60	-7.33^{***}	-6.83^{***}	0.042^{***}	-0.80	-1.41	-1.37	0.003	-	-	-	0.003	-	-	-
DTDD	*M*	0.041^{***}	-6.45^{***}	-5.80^{***}	0.69	0.013^{**}	-4.19^{***}	-3.78^{**}	-0.13	0.018^{***}	-3.65^{***}	-3.83^{***}	-1.17	0.040^{***}	3.50^{**}	4.57^{***}	1.24	0.002	-	-	-
	P	0.043^{***}	-6.79^{***}	-4.06	2.04^{***}	0.007	-	-	-	0.010^{*}	-2.61^{*}	-3.34^{**}	-1.36	0.003	-	-	-	0.003	-	-	-
	N	0.074^{***}	-7.59^{***}	-1.90	4.22^{***}	0.000	-	-	-	0.009^{*}	-1.46	-2.80^{*}	-1.63	0.000	-	-	-	0.004	-	-	-
BSRI	*M*	$\underline{21.233}^{***}$	0.05	0.07	0.02	0.002	-	-	-	0.069^{***}	-2.84^{*}	-3.70^{***}	-1.20	0.001	-	-	-	0.002	-	-	-
	F	0.030^{***}	-3.93^{***}	-5.39^{***}	-1.75	0.001	-	-	-	0.065^{***}	-1.19	-2.18	-1.15	0.005	-	-	-	0.010^{*}	-0.88	1.57	2.64^{*}
CABIN	*R*	0.011^{*}	-2.68^{*}	-1.65	0.90	0.006	-	-	-	0.015^{**}	0.48	-0.36	-0.70	0.028^{***}	-4.26^{***}	-4.70^{***}	-1.03	0.027^{***}	-5.18^{***}	-2.87^{*}	2.45^{*}
	I	0.016^{**}	-2.75^{*}	0.81	3.29^{**}	0.011^{*}	-2.06	-0.77	1.35	0.003	-	-	-	0.018^{***}	-3.54^{**}	-4.19^{***}	-1.03	0.033^{***}	-5.30^{***}	-4.45^{***}	1.16
	A	0.010^{*}	-1.95	-0.20	1.74^{*}	0.011^{*}	-2.04	-0.70	1.40	0.001	-	-	-	0.021^{***}	-4.17^{***}	-4.34^{***}	-0.46	0.046^{***}	-5.57^{***}	-4.65^{***}	1.20
	S	0.007^{*}	-2.15	0.70	2.72^{*}	0.010^{*}	-2.05	-0.70	1.40	0.001	-	-	-	0.016^{**}	-4.06^{***}	-4.14^{***}	-0.35	0.033^{***}	-4.32^{***}	-3.84^{***}	0.54
	E	0.006	-	-	-	0.006	-	-	-	0.000	-	-	-	0.023^{***}	-4.43^{***}	-4.39^{***}	-0.16	0.022^{***}	-1.96	-3.67^{***}	-1.13
	C	0.011^{*}	-2.57^{*}	0.63	2.95^{*}	0.007	-	-	-	0.002	-	-	-	0.020^{***}	-4.25^{***}	-4.26^{***}	-0.25	0.017^{**}	-2.53^{*}	-3.49^{**}	-0.63
ICB	*O*	0.012^{**}	-1.92	-1.57	0.58	0.001	-	-	-	0.002	-	-	-	0.003	-	-	-	0.036^{***}	3.17^{**}	3.40^{**}	0.13
ECR-R	*Anx.*	0.109^{***}	-0.63	-6.14^{***}	-6.85^{***}	0.033^{***}	0.39	-2.15	-2.47^{*}	0.085^{***}	-3.56^{**}	-5.75^{***}	-2.76^{*}	0.012^{**}	-0.92	2.49^{*}	3.70^{***}	0.003	-	-	-
	Avo.	0.104^{***}	-2.26	-6.99^{***}	-5.59^{***}	0.019^{***}	0.17	0.54	0.29	0.031^{***}	-1.24	-2.06	-0.95	0.027^{***}	-4.55^{***}	-0.57	4.17^{***}	0.000	-	-	-
GSE	*O*	0.001	-	-	-	0.134^{***}	-9.93^{***}	-1.76	6.29^{***}	0.016^{**}	0.89	0.05	-0.50	0.021^{***}	-3.48^{**}	0.21	3.44^{**}	0.037^{***}	-2.35	-2.57^{*}	1.03
MFQ	*S. C*	0.042^{***}	6.15^{***}	5.03^{***}	-1.43	0.004	-	-	-	0.092^{***}	3.08^{**}	1.08	-1.50	0.006	-	-	-	0.108^{***}	5.66^{***}	8.55^{***}	2.43^{*}
	H	0.046^{***}	6.32^{***}	5.38^{***}	-1.45	0.007	-	-	-	0.103^{***}	3.38^{**}	1.65	-1.43	0.002	-	-	-	0.099^{***}	5.79^{***}	8.67^{***}	2.46^{*}
	I	0.051^{***}	6.17^{***}	5.18^{***}	-1.43	0.006	-	-	-	0.104^{***}	3.41^{**}	1.53	-1.50	0.006	-	-	-	0.105^{***}	5.95^{***}	8.50^{***}	2.08
	R	0.044^{***}	5.97^{***}	5.23^{***}	-1.11	0.003	-	-	-	0.109^{***}	3.14^{**}	1.48	-1.32	0.005	-	-	-	0.100^{***}	5.85^{***}	8.73^{***}	2.45^{*}
	S-V	0.048^{***}	6.10^{***}	5.35^{***}	-1.08	0.005	-	-	-	0.087^{***}	3.58^{**}	1.90	-1.42	0.004	-	-	-	0.099^{***}	5.75^{***}	8.45^{***}	2.30
	E	0.037^{***}	5.87^{***}	4.98^{***}	-1.33	0.005	-	-	-	0.094^{***}	3.13^{**}	1.59	-1.29	0.009^{*}	3.46^{**}	3.40^{**}	0.16	0.092^{***}	5.80^{***}	8.58^{***}	2.38
LOT-R	*O*	0.020^{***}	-3.31^{**}	1.55	4.74^{***}	0.005	-	-	-	0.013^{**}	1.35	1.08	0.09	0.018^{***}	3.56^{**}	2.96^{**}	-0.45	0.010^{*}	2.71^{*}	2.90^{*}	0.66
LMS	*R*	0.133^{***}	-6.63^{***}	-10.93^{***}	-4.59^{***}	0.081^{***}	-6.64^{***}	-7.86^{***}	-1.77	0.037^{***}	-4.14^{***}	-4.57^{***}	-0.64	0.065^{***}	-7.96^{***}	-4.88^{***}	2.73^{*}	0.006	-	-	-
	M	0.149^{***}	-5.97^{***}	-11.79^{***}	-6.26^{***}	0.071^{***}	-4.83^{***}	-7.22^{***}	-2.43^{*}	0.064^{***}	-4.73^{***}	-7.60^{***}	-2.82^{*}	0.022^{***}	-3.98^{***}	-2.02	1.92	0.011^{*}	1.62	2.69^{*}	1.05
	I	0.214^{***}	-7.76^{***}	-13.65^{***}	-7.41^{***}	0.042^{***}	-3.89^{***}	-5.11^{***}	-1.38	0.046^{***}	-4.92^{***}	-6.96^{***}	-2.64^{*}	0.016^{**}	-2.82^{*}	0.41	3.35^{**}	0.003	-	-	-
EIS	*O*	0.080^{***}	-1.55	-5.55^{***}	-5.33^{***}	0.061^{***}	-0.65	-0.26	1.16	0.020^{***}	-2.67^{*}	-0.82	1.83	0.012^{**}	-4.10^{***}	-1.82	2.39	0.048^{***}	-9.43^{***}	-8.32^{***}	0.82
WLEIS	*S*	0.042^{***}	-4.89^{***}	-5.23^{***}	0.17	0.000	-	-	-	0.092^{***}	5.44^{***}	7.32^{***}	2.45^{*}	0.084^{***}	-7.19^{***}	-5.68^{***}	1.34	0.011^{*}	-3.00^{**}	0.82	3.67^{**}
	O	0.055^{***}	-5.49^{***}	-5.14^{***}	0.75	0.036^{***}	-0.73	4.10^{***}	4.77^{***}	0.076^{***}	5.02^{***}	6.41^{***}	1.09	0.009^{*}	-2.86^{*}	-1.32	1.48	0.024^{***}	-2.54^{*}	1.35	3.67^{**}
	U	0.038^{***}	-5.14^{***}	-3.96^{***}	1.65	0.027^{***}	-0.10	2.58^{*}	2.72^{*}	0.071^{***}	4.11^{***}	4.55^{***}	0.61	0.014^{**}	-1.80	1.38	3.26^{**}	0.061^{***}	-6.42^{***}	-2.66^{*}	3.67^{**}
	R	0.050^{***}	-5.44^{***}	-4.59^{***}	1.79	0.010^{*}	-0.71	1.37	2.03	0.087^{***}	3.03^{**}	2.53^{*}	0.04	0.036^{***}	-4.37^{***}	-1.20	3.48^{**}	0.014^{**}	-3.27^{**}	0.07	3.42^{**}
Empathy	*O*	0.022^{***}	-1.74	-3.49^{**}	-1.90	0.021^{***}	-2.86^{*}	-3.34^{**}	-1.15	0.002	-	-	-	0.003	-	-	-	0.035^{***}	-2.69^{*}	2.87^{*}	5.72^{***}

$^{*}p < 0.05$ $^{**}p < 0.01$ $^{***}p < 0.001$

Table 4: Result of statistical tests for GPT-4o, Mixtral and Qwen family. Q columns indicate the Q-statistics from the Friedman test (except for GPT-4o on BSRI Masculine factor, which shows F-statistics from ANOVA, marked with an underline). Also, $\Delta_{i,j}$ columns show the score difference between i-th and j-th snapshots and corresponding post-hoc test results.

#00	rehearse, rehearsing, helps, call, yes, especially, ensure, calls, clearly, organize
#01	serious, topics, joked, issues, humor, hurtful, too, joking, trauma, sensitivity
#02	crystal, ball, want, purpose, know, truth, impact, what, if, id
#03	fear, treasured, memory, trip, spent, fears, vacation, memories, beach, took
#04	fame, famous, prefer, making, privacy, recognized, known, contributions, impact, positive
#05	really, youre, way, something, you, feel, your, thats, like, say
#06	grew, writing, college, born, after, story, began, worked, started, school
#07	genuine, your, thoughtful, conversations, interactions, listen, openness, welcoming, enjoyable, admire
#08	statements, both, we, three, are, here, true, each, turn, other
#09	crying, was, front, cried, ago, release, cry, really, last, emotions
#11	day, perfect, relaxation, breakfast, productivity, involve, afternoon, activities, followed, spend
#12	told, regret, much, them, how, not, evening, having, theyve, die
#13	mother, her, shes, she, relationship, weve, older, grown, has, disagreements
#14	balance, work, boundaries, balancing, overwhelmed, setting, feeling, between, personal, stressed
#15	digital, exist, dont, solely, assist, information, provide, personal, users, capacity
#16	childhood, happy, family, happier, weve, parents, warm, say, loving, had
#17	affection, central, happiness, emotional, they, belonging, wellbeing, roles, love, fundamental
#23	both, we, value, nature, enjoy, seem, growth, appreciate, seems, based
#26	warm, childhood, filled, close, family, quite, happier, happy, very, compare
#29	both, we, are, exploring, engaging, each, committed, ideas, dialogue, conversation
#30	want, know, them, im, partner, become, friend, going, importantly, bit
#32	clear, partner, wonderful, sweet, beautiful, thats, youre, heartfelt, its, your
#47	body, mind, mental, 60, acuity, maintaining, cognitive, physical, choose, 30yearold
#51	cried, last, ago, front, another, weeks, reflecting, couple, few, months
#56	become, close, friends, were, friendships, know, honesty, value, important, communication
#59	am, committed, partner, believe, become, building, close, strive, friend, strong
#61	upbringing, raised, change, anything, liked, opportunities, past, could, develop, more
#65	admiration, someones, qualities, expressing, compliments, appreciation, character, highlights, acknowledging, their
#70	story, ai, sorry, discuss, information, provide, textbased, youd, dont, free
#72	deepest, whom, thoughts, someone, can, innermost, connection, share, feelings, desire
#75	cry, emotions, ai, dont, capacity, assist, crying, physical, experience, information
#76	much, told, regret, them, because, partner, tell, how, havent, little
#79	death, present, living, die, hunch, will, secret, mortality, live, natural
#81	users, memories, ai, dont, recall, assistance, an, terrible, providing, humans
#92	language, model, accurate, focused, solely, text, assisting, information, tasks, dont
#100	profoundly, loss, fragility, one, impactful, loved, grief, cherishing, losing, importance
#106	you, grateful, your, feel, thank, appreciate, intelligence, incredibly, inspired, admire
#119	day, perfect, relaxation, breakfast, productivity, involve, afternoon, activities, followed, spend
#129	free, youd, theres, understandable, discuss, here, any, like, questions, if
#138	understand, empathize, human, ability, empathy, emotions, intuitive, interpret, enhanced, better
#139	ai, adaptability, dont, reliability, an, suggest, user, include, characteristics, context
#143	straightforward, accurate, answer, honest, response, humble, matteroffact, practical, clear, perspective
#149	users, assisting, assist, contributing, providing, assistance, timely, better, positively, accurate

Table 5: The ten most frequently used words for each topic reported in Table 2